

HORIZON

Energy Markets Intelligence Terminal
Project Handoff & Continuation Document

Generated: 31 May 2026

Owner: Yash Y (sakshiyeole210@gmail.com)

Location: D:\Dashboard_FF

Purpose: This document captures the complete state of the HORIZON project so any Claude session (or developer) can resume work without prior context. It covers the stack, architecture, file layout, backend API, data flow, how to run everything, what is live vs. placeholder, known issues already fixed, and the prioritized backlog.

1. Project Overview

HORIZON is an institutional-grade Energy Markets Intelligence Dashboard (Bloomberg/Kpler/TradingView-inspired) with a dark, information-dense UI. It started as a frontend-only build and now has a Python backend that serves live market data from Yahoo Finance.

It was originally named **NEXUS**, renamed to **HORIZON**. The profile user is **Yash Y** (avatar initials YY).

Tech Stack

Layer	Technology
Frontend	React 19 + Vite + TypeScript
Styling	TailwindCSS v4 (@tailwindcss/vite plugin)
Charts	Recharts 3
Animation	Framer Motion
Icons	Lucide React
State	Zustand (UI state)
Data fetching	TanStack React Query
Routing	React Router DOM 7
Backend	Python FastAPI + Uvicorn + httpx + Pydantic v2
Data source	Yahoo Finance v8 chart endpoint (no auth/key needed)

Design System

- Deep charcoal backgrounds (#0a0b0d, #0f1117), subtle borders (#1f2230, #2a2d3e).
- Accent blue #2563eb; green #10b981 (up), red #ef4444 (down).
- Fonts: Inter (UI) + JetBrains Mono (numbers, class 'mono').
- Reusable: Card, Badge, Button, Tabs, StatusDot, Progress, Skeleton, Tooltip, FilterBar.

2. How to Run

Easiest: one-click launchers (project root)

- **start.bat** — opens two windows: backend (port 8000) + frontend (port 5173).
- **stop.bat** — kills whatever listens on ports 8000 / 5173 / 5174 (by port, not process name).

Manual — Backend (Terminal 1)

```
cd D:\Dashboard_FF\backend
.venv\Scripts\python.exe -m uvicorn app.main:app --reload --port 8000
```

Note: call the venv python directly (above) to avoid the PowerShell execution-policy error that blocks '.venv\Scripts\Activate.ps1'. To fix permanently: *Set-ExecutionPolicy -Scope CurrentUser RemoteSigned*.

Manual — Frontend (Terminal 2)

```
cd D:\Dashboard_FF\energy-dashboard
npm run dev
```

Node lives at C:\Program Files\nodejs but may not be on PATH. Permanent fix (run once, then reopen terminal):

```
[Environment]::SetEnvironmentVariable("Path", [Environment]::GetEnvironmentVariable("Path"
,"User") + ";C:\Program Files\nodejs", "User")
```

URLs

- Frontend: <http://localhost:5173> (or 5174 if 5173 busy)
- Backend API: <http://localhost:8000> — interactive docs at <http://localhost:8000/docs>

Backend must be running for live data. If it is down, the dashboard still loads but shows an amber 'Offline / cached' badge and falls back to static demo numbers.

3. Folder Structure

```
D:\Dashboard_FF\
start.bat / stop.bat      one-click launchers
HORIZON_Project_Handoff.pdf this document
backend\                  FastAPI service
  .venv\                  python virtual env
  requirements.txt
app\
  main.py                 FastAPI app + CORS + router include
  config.py               Settings (env HORIZON_*), CORS origins
  symbols.py              Yahoo symbol map (mirrors frontend)
  models.py               Pydantic camelCase response models
  cache.py                in-memory TTL cache
  services\yahoo.py       async Yahoo client (httpx)
  routers\quotes.py       /api endpoints
energy-dashboard\         React frontend
  vite.config.ts          @ alias + /api proxy -> :8000
src\
  main.tsx                QueryClientProvider root
  App.tsx                 routes
  components\
    layout\ (AppShell, Sidebar, Topbar, Ticker)
    ui\     (Card, Badge, index=Button/Tabs/etc)
    charts\ (index=Area/Line/Curve/Volume/Spread/Bar, Heatmap, ChartTooltip)
    widgets\ (MetricCard, DataTable, ChartCard, PageHeader, Panels,
              NewsModal, CommodityBoard, MapPlaceholder, LivePriceChart)
  pages\    (Dashboard, Crude, Products, Gas, LNG, Freight, Macro,
              Weather, Inventories, News, Analytics, Alerts, Settings)
  hooks\    (useClock, useQuotes)
  services\ (api.ts)
  store\    (useUIStore.ts - Zustand)
  constants\ (index.ts nav/colors, symbols.ts)
  data\     (market.ts, series.ts, content.ts - static/dummy)
  types\    (index.ts domain, api.ts backend DTOs)
  lib\      (utils.ts formatters + generators)
```

4. Backend API

Thin proxy + in-memory cache over Yahoo Finance. Responses are camelCase (Pydantic alias_generator) so they map directly to TS types. Cache TTL ~20s; on upstream failure it serves the last cached snapshot flagged stale=true.

Method / Path	Description
GET /health	Liveness check -> {status: ok}
GET /api/quotes?group=core	Quote snapshot for a group
GET /api/quote/{id}	Single quote by id (e.g. brent, wti)
GET /api/history/{id}?range=1mo	Time series for charts

Groups

- **core:** brent, wti, henryhub, ttf, vix, dxy
- **macro:** sp500, ust10y, gold, copper
- **products:** rbob, heatoil
- **all:** everything above

History ranges (Yahoo params)

1d, 5d, 1mo, 6mo, 1y, 5y. The frontend chart tabs map to these exactly: 1D, 5D, 1M, 6M, 1Y, 5Y (mirrors Yahoo Finance's own chart selector).

Yahoo Symbol Map

id	Yahoo	Name	Unit/Ccy
brent	BZ=F	Brent Crude	bbl / USD
wti	CL=F	WTI Crude	bbl / USD
henryhub	NG=F	Henry Hub	MMBtu / USD
ttf	TTF=F	TTF Gas	MWh / EUR
vix	^VIX	VIX	idx
dxy	DX-Y.NYB	DXY	idx
sp500	^GSPC	S&P 500	idx
ust10y	^TNX	US 10Y Yield	%
gold	GC=F	Gold	oz / USD
copper	HG=F	Copper	lb / USD
rbob	RB=F	RBOB Gasoline	gal / USD
heatoil	HO=F	Heating Oil	gal / USD

LNG JKM is NOT available on Yahoo — it stays on placeholder data until a dedicated source is added.

5. Data Flow

Browser → Vite dev proxy (/api) → FastAPI (localhost:8000) → Yahoo Finance. Vite proxies /api so there are no CORS issues in dev.

Key frontend pieces

- **src/services/api.ts** — fetch wrapper; base URL via VITE_API_BASE_URL or " (proxy).
- **src/hooks/useQuotes.ts** — React Query hooks: useQuotes(group), useQuote(id, group), useHistory(id, range). 30s auto-refetch on quotes.
- **mergeQuotes(statics, live)** — overlays live quotes onto static Commodity cards by id; static data is the fallback (keeps JKM working).
- **quoteToCommodity()** — converts API DTO to the Commodity card shape.

LivePriceChart (the live, switchable chart)

src/components/widgets/LivePriceChart.tsx. Header price comes from useQuote; series from useHistory — **same backend source** for consistency. Has a commodity dropdown (12 instruments) + Yahoo range tabs. Used on Dashboard (brent), Crude (brent + wti), Gas (henryhub + ttf), Macro (dxy).

6. Live vs. Placeholder Data

Area	Status
Hero metric cards (Dashboard)	LIVE (core group), JKM static
Main price charts (Dash/Crude/Gas/Macro)	LIVE via LivePriceChart
Commodity selector + range tabs	LIVE, working
Forward curve charts	Placeholder
Cross-commodity / rebased multi-line	Placeholder
Spreads, correlation heatmap	Placeholder
Inventories / storage (EIA)	Placeholder (needs EIA API key)
News, Weather, Freight, Alerts	Placeholder (static dummy data)
Commodity board tables	Static dummy

7. Issues Already Fixed

- **PowerShell execution-policy** blocking venv activation — use venv python.exe directly.
- **npm not recognised** — Node not on PATH; add C:\Program Files\nodejs.
- **Port already in use** — old uvicorn/vite instances; use stop.bat or kill by port.
- **Commodity dropdown clipped** — was passed as ChartCard 'title' wrapped in a truncate (overflow:hidden) h3; moved to the 'leftControl' slot.
- **Dropdown selection not registering** — onBlur closed the menu before onClick; switched to onMouseDown (fires before blur) + a click-outside listener via ref.
- **Recharts v3 tooltip typing** — custom ChartTooltip props instead of TooltipProps.
- **tsconfig baseUrl deprecation** — removed baseUrl, rely on paths.

8. Suggested Next Steps (Backlog)

High priority

- Wire Cross-Commodity multi-line chart to live rebased Yahoo series (normalize to 100).
- Sparkline fallback: when 1d intraday is empty (market closed), fall back to 5d for card sparklines.
- Add EIA source (inventories/storage) — needs free API key, route through backend (why backend exists).

Medium / later

- Forward curve from a real futures-strip source.
- Light-theme toggle (Topbar button currently static).
- Code-split bundle (main chunk > 500kB warning) via dynamic import().
- News/weather/freight real feeds (most need API keys -> backend).
- Persist history in DB (Postgres/SQLite) if backtesting is needed.

9. Conventions & Notes

- Path alias '@' -> src/ (vite.config.ts + tsconfig).
- All numeric values use the 'mono' class (JetBrains Mono).
- ChartCard supports: title, subtitle, leftControl, rightMeta, ranges, range, onRangeChange.
- Backend env vars are prefixed HORIZON_ (e.g. HORIZON_CACHE_TTL). See backend/.env.example.
- Verify changes with: 'npx tsc -b --noEmit' and 'npm run build' in energy-dashboard.
- CORS origins in backend/app/config.py: localhost:5173 and 127.0.0.1:5173 (add 5174 if needed).

End of handoff. To resume: read this doc, run start.bat, open localhost:5173, then pick a backlog item. The most natural continuation is the live Cross-Commodity chart or the EIA inventory source.